

Math 2

Partial Fractions:-

Case 2 factors of denominator are of 1st degree & some of them repeated.

* المقام يحتوي على عوامل من الدرجة الأولى بعضها مكرر.

* الشكل العام:

لو عندنا قوس مكرر في المقام زي $(c_3x + c_4)^2$ بنفكه بالتدرج (مرة أس 1 ومرة أس 2).

$$\frac{\boxed{}}{(c_1x + c_2)(c_3x + c_4)^2} = \frac{A}{c_1x + c_2} + \frac{B}{(c_3x + c_4)} + \frac{C}{(c_3x + c_4)^2}$$

Ex 0 Evaluate $I = \int \frac{x^3}{x^2 + 2x + 1} dx$

By using long division

$$\begin{array}{r} x-2 \\ x^2+2x+1 \overline{) x^3} \\ \underline{+x^3+2x^2+x} \\ -2x^2-x \\ \underline{+2x^2+4x+2} \\ 3x+2 \end{array}$$

$$\frac{x^3}{x^2+2x+1} = x-2 + \frac{3x+2}{x^2+2x+1} \quad (1)$$

$$\frac{3x+2}{x^2+2x+1} = \frac{3x+2}{(x+1)(x+1)} = \frac{3x+2}{(x+1)^2}$$

$$\frac{3x+2}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} \rightarrow \text{multiply both sides by } (x+1)^2$$

$$3x+2 = A(x+1) + B$$

Put $x = -1$

$$3(-1) + 2 = 0 + B \Rightarrow B = -1$$

$$3x+2 = A(x+1) - 1$$

Put $x = 0$ نعوض بأي قيمة

$$0+2 = A(0+1) - 1 \Rightarrow A = 3$$

$$\frac{3x+2}{(x+1)^2} = \frac{3}{(x+1)} + \frac{-1}{(x+1)^2}$$

Substitute into (1)

$$\frac{x^3}{x^2+2x+1} = x-2 + \frac{3}{(x+1)} - \frac{1}{(x+1)^2}$$

$$\therefore \int \frac{x^3}{x^2+2x+1} dx = \int \left(x-2 + \frac{3}{(x+1)} - \frac{1}{(x+1)^2} \right) dx$$

$$= \int x dx - \int 2 dx + 3 \int \frac{1}{(x+1)} dx - \int (x+1)^{-2} dx$$

$$= \frac{x^2}{2} - 2x + 3 \ln(x+1) - \frac{(x+1)^{-1}}{-1} + C$$

EX② Evaluate $I = \int \frac{x^4 - x^3 - x - 1}{x^3 - x^2(x-1)} dx$

By using long division

$$\begin{array}{r} x \\ x^3 - x^2 \overline{) x^4 - x^3 - x - 1} \\ \underline{x^4 - x^3} \\ -x - 1 \end{array}$$

$$\frac{x^4 - x^3 - x - 1}{x^3 - x^2} = x + \frac{-x-1}{x^3 - x^2}$$

$$\frac{-x-1}{x^3 - x^2} = \frac{-x-1}{x^2(x-1)}$$

عامل من الدرجة الأولى متكرر

$$\frac{-x-1}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{(x-1)}$$

multiply both sides by $x^2(x-1)$

$$-x-1 = Ax(x-1) + B(x-1) + Cx^2$$

Put $x=1$

$$-1-1 = 0+0+C \Rightarrow C = -2$$

Put $x=0$

$$0-1 = 0+B(0-1)+0$$

$$-1 = -B \Rightarrow B = 1$$

Put $x=2$

$$-2-1 = A(2)(2-1) + (1)(2-1) + (-2)(4)$$

$$-3 = 2A + 1 - 8$$

$$-3+7 = 2A \Rightarrow A = 2$$

$$\frac{-x-1}{x^2(x-1)} = \frac{2}{x} + \frac{1}{x^2} + \frac{-2}{x-1}$$

Substitute into ①

$$\frac{x^4 - x^3 - x - 1}{x^3 - x^2} = x + \frac{2}{x} + \frac{1}{x^2} - \frac{2}{x-1}$$

$$\therefore \int \frac{x^4 - x^3 - x - 1}{x^3 - x^2} dx = \int x dx + 2 \int \frac{1}{x} dx + \int x^{-2} dx - 2 \int \frac{1}{x-1} dx$$

$$= \frac{x^2}{2} + 2 \ln x + \frac{x^{-1}}{-1} - 2 \ln(x-1) + C$$

Case ③ Denominator contains factors of 2nd degree Not repeated. $(1-0)8 + 0 = 1-0$

* المقام يحتوى على عوامل من الدرجة الثانية غير مكررة

* الشكل العام:

لو المقام فيه معادلة تربيعية لا تحلل (زى $X^2 + 4$) بنفرض البسط يتاعى كمعادلة خطية $(CX + D)$

$$\frac{\boxed{}}{(C_1X^2 + C_2)(C_3X^2 + C_4)} = \frac{AX + B}{C_1X^2 + C_2} + \frac{CX + D}{C_3X^2 + C_4}$$

EX ③ Express $\frac{2X^3 - 4X - 8}{X(X-1)(X^2+4)}$ $\rightarrow$ درجة البسط 3 $\rightarrow$ درجة المقام 4

using Partial Fraction

$$\frac{2X^3 - 4X - 8}{X(X-1)(X^2+4)} = \frac{A}{X} + \frac{B}{X-1} + \frac{CX + D}{X^2+4}$$

multiply both sides by $X(X-1)(X^2+4)$

$$2X^3 - 4X - 8 = A(X-1)(X^2+4) + BX(X^2+4) + (CX+D)X(X-1)$$

Put $X=0$

$$0 - 0 - 8 = A(0-1)(0+4) + 0 + 0$$

$$-8 = -4A \Rightarrow A = 2$$

Put $X=1$

$$2 - 4 - 8 = 0 + B(1)(1+4) + 0$$

$$-10 = 5B \Rightarrow B = -2$$

Put $X=-1$

$$2(-1)^3 - 4(-1) - 8 = A(-1-1)((-1)^2+4) + B(-1)(1+4) +$$

$$(-C+D)(-1)(-1-1)$$

$$-2 + 4 - 8 = A(-10) + B(-5) + (-C+D)(2)$$

$$-6 = 2(-10) + (-2)(-5) - 2C + 2D$$

$$-6 = -20 + 10 - 2C + 2D$$

$$-2C + 2D = 4 \quad (1)$$

Put $X=2$ عوض بيها عوض بيها

$$16 - 8 - 8 = A(8) + B(16) + (2C+D)(2)$$

$$0 = 16 - 32 + 4C + 2D$$

$$4C + 2D = 16 \quad (2)$$

$$(2) - (1) \Rightarrow 6C = 12 \Rightarrow C = 2$$

Substitute into (1) $\Rightarrow -4 + 2D = 4$

$$D = 4$$